Uses of Moore-Penrose Pseudoinverse

The Constrained Motion web application uses Moore-Penrose pseudoinverses to compute machine dynamics; however, MPI calculation is buried deep in the code. To get familiar with underlying concept of MPI, one can examine more mundane uses.

In two scenarios, I pretend to be processing measurement data dealing with two elementary circuits. We set voltages (V1, V2 ...) and measure current Isum with the purpose to determine the unknown resistances (actually, conductances, which are reciprocals to resistances). Labels V1, V2, V3, Vmeas, R1, R2, R3 at component tops are circuit component identificators; labels V1, V2, V3, 0, g1, g2, g3 at the bottom of components are the nominals (values) of these components. Sorry for V1, V2, V3 playing two roles in schematic drawings, a convention familiar to electrical engineers.

The first circuit sums up currents from three branches; assume that for some reason, you are allowed to take only two measurements. In each measurement, you set new voltage values for three constant voltage sources V1, V2, and V3. The sum of branch currents is the current through the measurement voltage source Vmeas. The voltage across this source is set to zero, the typical trick used to measure currents in simulators.

The second circuit sums up currents from two branches; this time, you are allowed to make three measurements, changing voltages you set for two constant voltage sources V1 and V2. The sum of two branch currents is the current through the measurement voltage source Vmeas.

Detailed explanation of circuit component definitions is given in Fig. 2. Resistor conductances are 'gn' (= 1/Rn), branch currents are 'In' (= Vn · gn).

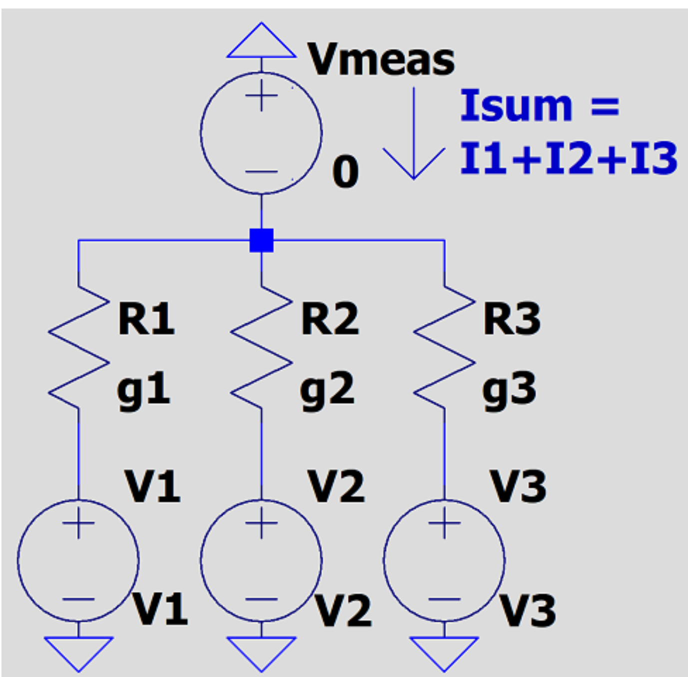

Fig. 1 Three branches, two measurements

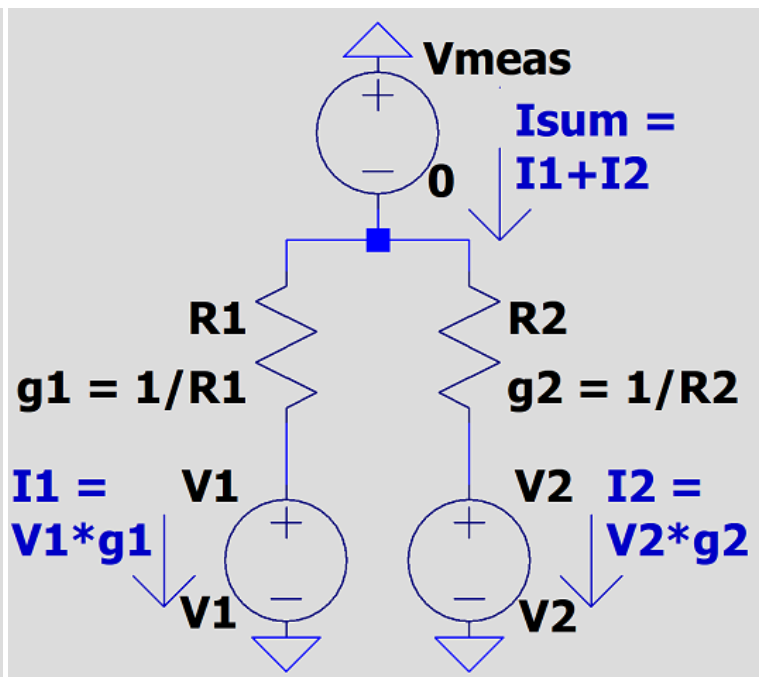

Fig. 2 Two branches, three measurements

Two measurement data sets are available in Scenario 1. The first measurement has voltage source values set as in array [V1:2, V2:1, V3:0], with the current through Vmeas Isum = 1; the second, with [V1:0, V2:1, V3:2] and Isum = 1. This gives us a linear system of two equations for three unknown variables [g1, g2, g3]:

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} g1 \\ g2 \\ g3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The convention is to write this system in the form

$A \cdot g = b$

As matrix A is not square, we cannot expect to find a solution in the form $g = A^{-1} \cdot b$. It is where MPI comes to the rescue:

$g = A^{+} \cdot b$

For linearly independent rows (as is Scenario 1 case), $A{\cdot}A^{*}$ is invertible and MPI can be computed as

$A^{+} = A^{*}{\cdot}(A{\cdot}A^{*})^{-1}$ /*Wikipedia, textbooks, course notes*/

A^{*} is a transposed matrix with complex conjugated elements. Our matrix A is real, therefore, $A^{*} = A^{T}$.

However, $g = A^{+} \cdot b$ is not a unique solution for Scenario 1. It is easy to verify that

$g = A^{+} \cdot b + [I - A^{+}{\cdot}A]{\cdot}w$

where w is an arbitrary vector [w1, w2, w3], I is an identity matrix, also solves the system $A \cdot g = b$. Left multiply $g = A^{+} \cdot b + [I - A^{+}{\cdot}A]{\cdot}w$ by A, and you are back to the source equation:

$A \cdot g = A{\cdot}A^{+}{\cdot}b + [A - A{\cdot}A^{+}{\cdot}A]{\cdot}w = b + [A - A]{\cdot}w$

For completeness, I give an example, writing down numerical values into above expressions using a programmer-oriented linear notation for matrices.

A = [[2, 1, 0], [0, 1, 2]]

$A^{*} = A^{T}$ = [[2, 0], [1, 1], [0, 2]]

$A{\cdot}A^{*}$ = [[5, 1], [1, 5]]

$(A{\cdot}A^{*})^{-1}$ = [[5/24, -1/24], [-1/24, 5/24]]

$A^{+} = A^{*}{\cdot}(A{\cdot}A^{*})^{-1}$ = [[5/12, -1/12], [1/6, 1/6], [-1/12, 5/12]]

$g = A^{+} \cdot b + [I - A^{+}{\cdot}A]{\cdot}w$ = [1/3, 1/3, 1/3] + [[1/6, -1/3, 1/6], [-1/3, 2/3, -1/3], [1/6, -1/3, 1/6]]·[[w1], [w2], [w3]]

We have that g = [1/3, 1/3, 1/3] is a solution, but adding [[1, -2, 1], [-2, 4, -2], [1, -2, 1]]·[[w1*6], [w2*6], [w3*6]] with arbitrary [[w1], [w2], [w3]] also gives a solution. Is there any practical use of expressing solution in this form? Some certainly exists, because although [[w1], [w2], [w3]] can be chosen arbitrarily, the components of vector [[1, -2, 1], [-2, 4, -2], [1, -2, 1]]·[[w1*6], [w2*6], [w3*6]] are still linearly dependent. What's more, a physics of your problem can provide additional contraint, such as, for example, non-negativity of conductance (resistance) values.

In linear algebra parlance, Scenario 1 generates an underdetermined system of linear equations.

Three measurement data sets are available in Scenario 2. The first measurement is taken with voltage source values set as in array [V1:2, V2:1], with the current through Vmeas Isum = 2; the second, with [V1:1, V2:2] and Isum = 1, and the third, with [V1:2, V2:-2] and Isum = -2. This gives us a linear system of three equations for two unknowns [g1, g2]:

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 2 & -2 \end{pmatrix} \cdot \begin{pmatrix} g1 \\ g2 \\ \ \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

As in the first scenario, we write down this system in the form

$A \cdot g = b$

This time, we have three measurement data sets to find just two unknown variables. The system is overdetermined. Overdetermined systems are almost always inconsistent and have no exact solution. In an idealized experiment with two branch currents, maximum two equations can be linearly independent, and if all the equations happen to become linearly dependent, you cannot find unique g1, g2 values -- at least two equations must be linearly independent (but if three or more equations are linearly independent, the system with two unknowns becomes inconsistent). In reality, measurement errors almost always make all the equations linearly independent and, under some assumption about error statistics, we can only estimate the most probable g1, g2 values (strictly speaking, corresponding to a maximum of probability density function). Sometimes the only information at our disposal is the measured data and then we are using the least squares approximation to minimize the sum of squared 'residuals', the differences between observed values (currents) in 'b' and the values given by the right part of the equation (A · g). Mathematically speaking, we aim to minimize

|| A · g - b ||

over all possible 'g' values.

Deciphering || ... || notation for our linear system, we have to minimize

$(2\cdot g1 + g2 - 2)^2 + (g1 + 2\cdot g2 - 1)^2 + (2\cdot g1 - 2\cdot g2 + 2)^2$

that is, to find a vector [g1, g2] that minimizes the above expression.

Symbolically, the solution to a 'minimize || A · g - b ||' problem is the 'normal equation' so called because this minimization can be interpreted as a geometrical problem. You can learn about the 'normal equation' in textbooks or linear algebra course notes, e.g., [CME 302 Numerical Linear Algebra]

$A^T\cdot A\cdot g = A^T\cdot b$ /* normal equation, A^T is the transpose of A */

Substituting numerical values,

$A^T\cdot A$ = [[9, 0], [2, 7]]

$A^T\cdot b$ = [1, 6]

solving equation

```
  /      \      /   \     / \
 | 9   0 |     | g1 |    | 1 |
 | 2   7 |  ·  | g2 | =  | 6 |
  \      /      \   /     \ /
```

we have g1 = 1/9, g2 = 52/63.

To solve a general least-square problem and for coding purposes, one can use the Moore Penrose pseudoinverse, likewise it is used to find a general solution for the underdetermined linear system.

Similar to how we do it for an underdetermined system, the expression

$g = A^+ \cdot b$

is the solution that minimizes || A · g - b || (see prove elsewhere) and this time we should not add the $[I - A^+\cdot A]\cdot w$ term.

As we have now linearly independent columns, invertible is the $A^*\cdot A$ product and MPI can be computed as

$A^+ = (A^*\cdot A)^{-1}\cdot A^*$

Substituting values,

$(A^*\cdot A)$ = [[9, 0], [2, 7]]

$(A^*\cdot A)^{-1}$ = [[1/9, 0], [-2/63, 1/7]]

$(A^*\cdot A)^{-1}\cdot A^*$ = [[2/9, 1/9, 2/9], [5/63, 16/63, -13/63]]

$g = A^+ \cdot$ [[2], [1], [-2]] = [1/9, 52/63]

The result is identical with that obtained with direct solving of equations.

The sum of all residuals with this solution is

$(2 \cdot g1 + g2 - 2)^2 + (g1 + 2 \cdot g2 - 1)^2 + (2 \cdot g1 - 2 \cdot g2 + 2)^2 = (-60/63)^2 + (48/63)^2 + (81/63)^2 \sim= 3.14$

A square root of this value is 1.77, much greater then computed conductances. The measurement precision is rather poor.